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Tech Job Skills Analysis in Saudi Arabia

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1 Introduction

The job market is undergoing continuous change, particularly within technology-driven sectors, where evolving digital and analytical skill requirements make it increasingly difficult for job seekers, educators, and policymakers to identify which competencies are currently most in demand. In Saudi Arabia, understanding skill demand in technology-related roles is especially important for aligning workforce capabilities with the needs of a rapidly growing digital economy. Online job postings for data, artificial intelligence, and related technology positions provide a rich and up-to-date source of information on employer skill requirements; however, this information is not always systematically analyzed to extract actionable insights. Accordingly, this project seeks to address this gap by answering the following research question: What skills are most in demand in technology-focused job roles in the Saudi job market based on online job postings?

2 Literature Review

2.1 Skill Categorization Methods in Job Advertisements [1]

⑩ Problem

The study addressed the problem of identifying the most effective method for extracting and categorizing skill requirements from online job advertisements. Given the variety of existing approaches, the authors aimed to evaluate which method best captures economically meaningful skill information by measuring its explanatory power in wage regressions.

⑩ Dataset

The study used a large-scale dataset of over one million online job advertisements from the UK labor market. The dataset included job titles, detailed job descriptions, and wage information. These advertisements provided unstructured textual data from which skill requirements were extracted and later analyzed in relation to wage variation.

⑩ Methods

The study compared two main approaches for skill extraction: a top-down method based on predefined skill dictionaries and a bottom-up approach using Latent Dirichlet Allocation (LDA) topic modeling. The extracted skills were incorporated into wage regression models to evaluate their explanatory power. The effectiveness of each method was assessed based on its ability to explain wage variation.

⑩ Findings

The study found that the bottom-up LDA-based method outperformed the top-down dictionary approach in explaining wage variation. The LDA method demonstrated higher explanatory power in wage regression models, suggesting that unsupervised skill extraction techniques capture economically meaningful skill patterns more effectively. The results also confirmed that online job advertisements serve as a reliable source for analyzing labor market skill requirements.

2.2 An investigation of skill requirements for business and data analytics positions: A content analysis of job advertisements [2]

⑩ Problem

The study investigates the skill requirements for business and data analytics job positions in the United States. Specifically, it aims to clarify the differences and similarities in required skill sets across four analytics-related job categories: Business Analyst (BA), Business Intelligence Analyst (BIA), Data Analyst (DA), and Data Scientist (DS). The study addresses the lack of clear classification and structured mapping of skills associated with these roles in the analytics domain

⑩ Dataset

The dataset consisted of 1,235 online job postings collected from Indeed.com over a three-month period (December 2016 to February 2017). The study focused on job postings from four U.S. states: Arkansas, Florida, Kansas, and Missouri. Web scraping and the Indeed API were used to collect full job descriptions for content analysis

⑩ Methods

The authors applied content analysis as the primary method. They developed a classification framework consisting of multiple skill categories (e.g., decision-making, programming, statistics, structured data management, communication, organization).

Using text processing techniques, they extracted unigrams, bigrams, and trigrams from job descriptions and matched them against predefined skill keywords. The study then calculated the frequency of skills and skill categories across job postings. Additionally, pairwise comparisons were conducted between BA and BIA, and between DA and DS.

⑩ Findings

The findings indicate that decision-making skills are the most demanded across all four job categories. However, differences were observed in technical skill requirements:

- Data Scientist positions require stronger programming and statistical skills.

- Data Analyst roles emphasize structured data management and statistics.
- Business Analyst roles focus more on domain knowledge and soft skills.
- Business Intelligence Analyst roles require a combination of structured data management and statistical tools.

The study concludes that Data Scientist positions are the most technically demanding, while Business Analyst roles rely more heavily on business domain knowledge and communication skills.

2.3 Demand for Skills on the Labor Market in the IT Sector [3]

⑩ Problem

The study addressed the challenge of identifying key skillsets demanded by employers in the IT sector. Given the rapid evolution of technical and soft skill requirements, the authors aimed to develop a systematic method for extracting and standardizing skill information from unstructured online job advertisements. The study specifically sought to determine which skills and combinations of skills are most demanded across different IT occupational groups.

⑩ Dataset

The study used a large dataset of online job vacancies obtained from the HeadHunter API, the largest recruitment platform in Russia. The dataset focused on IT-related job postings in Saint Petersburg between 2015 and 2019. After filtering and regrouping, approximately 56,000 IT vacancies were analyzed. The dataset included structured information (e.g., vacancy IDs and specialization codes) and unstructured text fields such as job descriptions and listed key skills.

⑩ Methods

The study applied Natural Language Processing (NLP) techniques to extract and standardize skills from unstructured job advertisements. The methods included TF-IDF, tokenization, and the construction of uni-grams, bi-grams, and tri-grams. Skills were standardized using manual synonym matching and similarity measures. Jaccard similarity was used to identify and match related skill terms and to extract significant pairs and triplets of skills across occupational groups.

⑩ Findings

The study identified 435 standardized skills extracted from unstructured IT job advertisements and revealed the most frequently demanded technical and non-technical skills in the sector, such as HTML/CSS, JavaScript, SQL, and Python. Beyond individual skills, the study successfully extracted occupation-specific skill combinations (pairs and triplets) that differentiate between IT occupational groups. The results demonstrated that certain skillsets strongly characterize specific job categories, and that combining similarity measures with n-gram analysis enables effective standardization and classification of skills. The findings confirm that online job advertisements provide a reliable source for identifying both core and occupation-specific skill demand in the IT labor market.

2.4 Organizational and End-User Information Systems Job Market: An Analysis of Job Types and Skill Requirements [4]

⑩ Problem

The study addresses the lack of systematic understanding regarding the structure of the Organizational and End-User Information Systems (OEIS) job market. While substantial research had examined traditional technical Information Systems (IS) roles, there was limited empirical evidence identifying the specific job categories, skill requirements, and responsibility profiles associated with OEIS positions. The authors sought to determine how the OEIS job market is structured, what types of positions exist within it, and what competencies employers explicitly demand in job advertisements. The study aimed to provide clarity for curriculum design, career planning, and alignment between academic programs and labor market expectations.

⑩ Dataset

The dataset consisted of 484 job advertisements collected from U.S.-based online employment sources. These advertisements specifically targeted Organizational and End-User Information Systems roles rather than purely technical programming or software engineering positions. Each advertisement was treated as a unit of analysis. The dataset included information about job titles, responsibilities, required skills, educational background, work experience expectations, and company characteristics

⑩ **Methods**

The study employed a content analysis methodology to systematically examine the job advertisements. The authors developed a classification framework to categorize positions into job types and subcategories. Through iterative coding and refinement, job titles and descriptions were analyzed to identify recurring patterns in responsibilities and skill requirements. Positions were grouped into broader categories and further subdivided into more specific job classifications. Quantitative frequency analysis was then applied to determine the relative demand for each job type and associated skills. The analysis also examined experience requirements and organizational characteristics to provide contextual understanding of market demand.

⑩ **Findings**

The study identified five main OEIS job categories, End-User Support, Business Analyst, Training, Web and Interface Design, and Technical Writing, divided into 24 subcategories. End-User Support and Business Analyst roles were most common. Employers emphasized technical, communication, analytical, and business skills, and many positions required prior experience. Overall, the findings highlight a diverse, non-programming-focused job market requiring interdisciplinary competencies.

2.5 Literature Comparison

Across the reviewed studies, a clear commonality is the use of online job advertisements as the primary data source for analyzing labor market skill demand. All four studies rely on unstructured textual data from job descriptions and titles, demonstrating the value of digital job platforms for extracting employer-required skills. Despite differences in geographic focus (UK, United States, and Russia) the studies consistently confirm that online job postings provide rich and scalable data for labor market analysis. However, the studies differ significantly in their analytical focus and modeling approaches. Study [1] evaluates competing skill extraction methodologies by comparing dictionary-based methods with LDA topic modeling and validating results using wage regression. Study [2] applies structured content analysis and predefined skill frameworks to compare analytics-related job roles. Study [3] employs NLP techniques such as TF-IDF, n-grams, and similarity measures to standardize and cluster IT skills and identify occupation-specific skill combinations. In contrast, Study [4] focuses primarily on developing a classification framework to structure the OEIS job market, using iterative content analysis and frequency analysis rather than advanced computational modeling. This reflects a shift from methodological optimization and skill extraction toward market structure and job taxonomy development.

While the reviewed studies provide valuable insights into skill extraction and labor market analysis, several limitations remain. Many studies are geographically limited and rely on predefined skill taxonomies, which may restrict the identification of emerging skills. Additionally, the Saudi Arabian labor market particularly in technology and data-related roles has not been sufficiently explored. This highlights the need for localized analysis to better understand employer skill requirements in Saudi Arabia's growing data and AI job market.

3 Data Source:

Job postings data was collected using the JSearch API provided via RapidAPI. The dataset consists of technology-related job advertisements published on multiple job platforms (such as LinkedIn and other job boards) and filtered specifically for positions located in Saudi Arabia. The collected data focuses on roles within the data and AI domains, including Data Scientist, Data Analyst, and related positions.

- **Data Collection:**
The data was collected by sending HTTP GET requests to the JSearch API endpoints using Python. Specific query parameters were used to retrieve job postings related to data and AI roles within Saudi Arabia. The API responses were returned in JSON format and then converted into a structured tabular format for further analysis. Multiple pages of results were fetched to increase the number of job postings collected.

```
response = requests.get(url, headers=headers, params=params)
data = response.json()
```

3.1 Data Description

1. Number of Observations: 1025
2. Number of features: 31
3. Data Types:

Feature	Data Type	Description
job_id	Qualitative (Nominal)	Unique identifier for each job posting.
job-title	Qualitative (Nominal)	Title of the advertised job position.
employer_name	Qualitative (Nominal)	The name of the company offering the job.
employer_logo	Qualitative (Nominal)	Link to the employer's logo image.
employer_website	Qualitative (Nominal)	Official website of the employer.
job_publisher	Qualitative (Nominal)	Platform where the job was published.
job_employment_type	Qualitative (Nominal)	Primary employment type of the job.
job_employment_types	Qualitative (Nominal)	All employment types associated with the job.

job_apply_link	Qualitative (Nominal)	URL used to apply for the job.
job_apply_is_direct	Qualitative (Nominal)	Indicates whether the application is direct on the platform or via an external website.
job_city	Qualitative (Nominal)	City where the job is located.
job_country	Qualitative (Nominal)	Country where the job is located.
job_latitude	Quantitative (Ratio)	Latitude of the job location.
job_longitude	Quantitative (Ratio)	Longitude of the job location.
job_description	Qualitative (Nominal)	Textual description of the job role.
job_benefits	Qualitative (Nominal)	Benefits offered with the job.
job_google_link	Qualitative (Nominal)	Google Jobs link for the posting.
job_salary	Quantitative (Ratio)	Text-based salary information.

job_min_salary	Quantitative (Ratio)	Minimum salary offered.
job_max_salary	Quantitative (Ratio)	Maximum salary offered.
job_salary_period	Qualitative (Nominal)	Salary payment period.
job_onet_soc	Qualitative (Nominal)	O*NET occupational classification code.
job_onet_job_zone	Quantitative (Ordinal)	Experience and preparation level indicator.
search_query	Qualitative (Nominal)	Search keyword used to retrieve the job.
job_posted_at_datetime_utc	Quantitative (Interval)	Job posting date and time (UTC).
job_posted_at_timestamp	Quantitative (Ratio)	Timestamp of job posting.
job_offer_expiration_datetime_utc	Quantitative (Interval)	Job offer expiration date (UTC).
job_offer_expiration_timestamp	Quantitative (Ratio)	Timestamp of job offer expiration.
job_required_experience	Qualitative (Ordinal)	Required experience level.

job_required_skills	Qualitative (Nominal)	Required or preferred skills.
job_required_education	Qualitative (Ordinal)	Required education level.

3.2 Dataset Bias Evaluation:

⑩ Representation Bias:

The dataset is collected from online job platforms through the JSearch API, which may overrepresent technology roles from large companies and urban areas while underrepresenting smaller organizations or roles advertised through informal or offline channels.

⑩ Measurement Bias:

Some variables, such as salary information and required skills, are inconsistently reported across job postings. In addition, skill requirements are extracted from textual descriptions, which may vary in detail between employers.

⑩ Historical Bias:

The dataset may reflect existing inequalities in the job market, such as greater visibility of roles requiring English proficiency or advanced technical backgrounds, potentially limiting representation of certain groups or regions.

4 Objectives

- To identify the most frequently demanded skills in data and AI-related job postings in Saudi Arabia.
- To examine the distribution of different skill categories, such as technical, analytical, and soft skills, within Saudi data and AI job advertisements.
- To analyze variations in skill requirements across different data and AI roles, including positions such as Data Scientist and Data Analyst.
- To explore common combinations of skills requested by employers in data and AI job postings in the Saudi job market.

5 Methods

1. Data Preprocessing

First, duplicate entries were identified and removed to prevent inflation of skill frequency counts and role distribution analysis. Duplicates existed both at the record identifier level and at the content level (repeated postings with identical core information). Retaining only unique postings ensured analytical validity. Second, missing values were evaluated at the attribute level. Fully empty attributes were excluded from analysis, while partially missing information was retained when relevant to the study objectives to preserve dataset size and representativeness.

Third, textual cleaning was applied to remove non-informative artifacts introduced during web scraping and to standardize text formatting. Consistent casing and normalization were applied to ensure uniform representation of English skill terms.

Finally, terminology normalization was performed to address variations in how the same skills were written. Equivalent expressions were unified under standardized labels before frequency computation to avoid fragmented counts and inaccurate demand estimation.

2. Data Transformation:

The transformation process focuses on organizing the dataset into structured and analyzable formats aligned with the research objectives.

Skills will be standardized to unify different expressions of the same concept to ensure accurate frequency measurement and pattern detection.

Job roles will be grouped into consistent categories to enable comparison of skill requirements across different data and AI positions.

Skill categories will classify extracted skills into broader groups such as technical, analytical, and soft skills to support proportional distribution analysis.

Finally, the structured dataset will be stored in formats such as CSV or JSON to ensure compatibility with analytical and data mining tools.

3. Analysis Techniques:

Once the data is preprocessed and transformed, a set of analytical techniques will be applied to achieve the project objectives related to skill demand in data and AI job postings in Saudi Arabia.

Skill frequency analysis will be used to extract skills from job descriptions and required skills fields and count their occurrences across job postings. This technique directly supports identifying the most frequently demanded skills in the Saudi job market.

Skill category distribution analysis will be conducted by grouping extracted skills into technical, analytical, and soft skill categories and calculating their relative proportions. This analysis supports understanding the balance of different skill types requested by employers.

Role-based comparative analysis will analyze differences in skill requirements across data and AI roles by grouping job postings based on job titles such as Data Scientist and Data Analyst. This technique supports examining variations in skill demand across roles.

Skill co-occurrence analysis will be applied to identify common combinations of skills that appear together within the same job postings. This analysis supports exploring how employers bundle skills when advertising data and AI positions.

4. Reporting insights:

The analysis results will be reported in a clear and structured manner to address the project objectives related to skill demand in data and AI job postings in Saudi Arabia.

Frequently demanded skills will be presented using ranked summaries to highlight the most in-demand competencies. The distribution of skill categories will be summarized to show the balance between technical, analytical, and soft skills. Variations in skill requirements across different data and AI roles will be reported through comparative summaries. In addition, common combinations of skills requested by employers will be highlighted to reflect typical skill sets in job postings.

Overall insights will be summarized in relation to the Saudi job market context.

6 Challenges Faced and Recommendations

During the data collection phase, several challenges were encountered when using the JSearch API. The API subscription was limited, which restricted the number of requests and the amount of data that could be collected. In addition, the data retrieval process was time-consuming due to API response time and pagination when fetching multiple job postings.

For future work, efforts could focus on expanding data sources to reduce platform-specific bias and improve market representativeness. Improving cross-platform deduplication techniques would help remove duplicate postings across different platforms. Furthermore, refining skill extraction methods could enhance data accuracy and strengthen the reliability of the analytical results.

7 Primary Data Analysis

7.1 Primary Data Findings

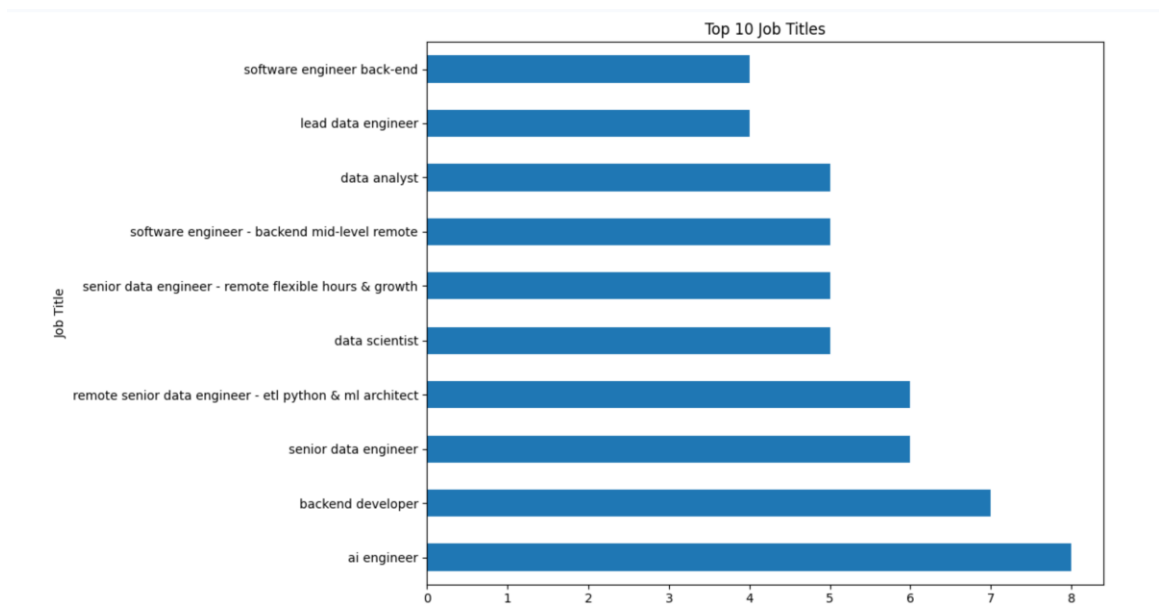
This section presents the findings derived from the primary data analysis. Exploratory Data Analysis was conducted to identify patterns, trends, and relationships within the dataset of 1,020 job postings. The following subsections summarize the key insights obtained, supported by detailed visualizations and descriptive statistics.

7.1.1 Job Titles Distribution

The initial EDA reveals a strong concentration of demand in high level technical engineering and data focused roles.

- **AI Engineer** emerged as the most frequent role , indicating a significant push toward artificial intelligence integration in the local market.
- **Backend Developer** and various **Senior Data Engineer** roles also represent a high volume of the dataset.
- The prevalence of roles like **Data Scientist** and **Data Analyst** confirms that organizations are actively seeking professionals to bridge the gap between raw data collection and actionable business intelligence.

The distribution suggests that while specialized Data roles are prominent, there is a substantial overlap with core software engineering, particularly in backend systems and infrastructure, which are essential for supporting AI and data pipelines.

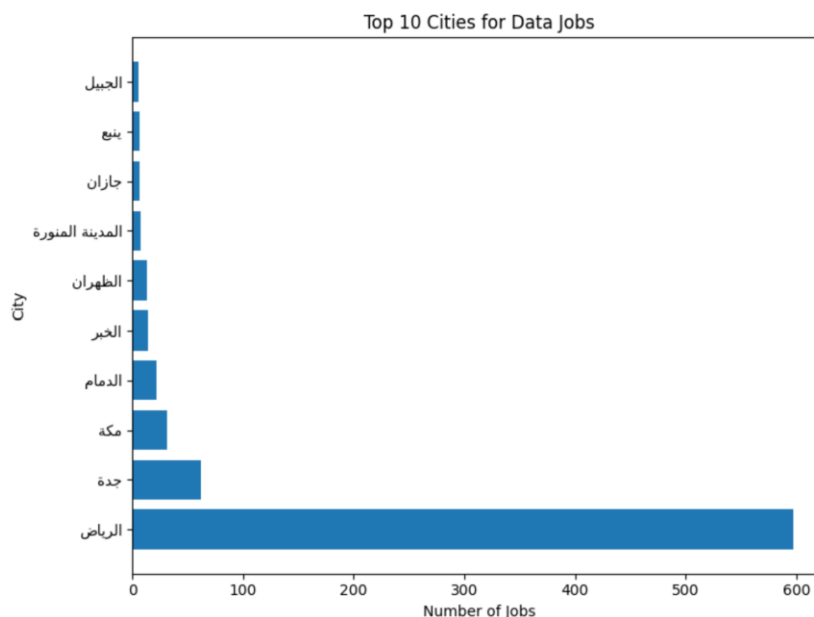


7.1.2 City Distribution

When looking at the locations of these job postings, a clear pattern emerges. Riyadh appears to be the primary hub for data and technology related employment in Saudi

Arabia, accounting for approximately 600 postings in the dataset. This concentration is expected, as Riyadh hosts the headquarters of many government entities and major companies.

In contrast, other cities such as Jeddah and Dammam show considerably fewer opportunities. For example, Jeddah records only around 50 job postings. These findings suggest that individuals seeking careers in Data Science or Artificial Intelligence are more likely to encounter greater employment opportunities in the capital city.



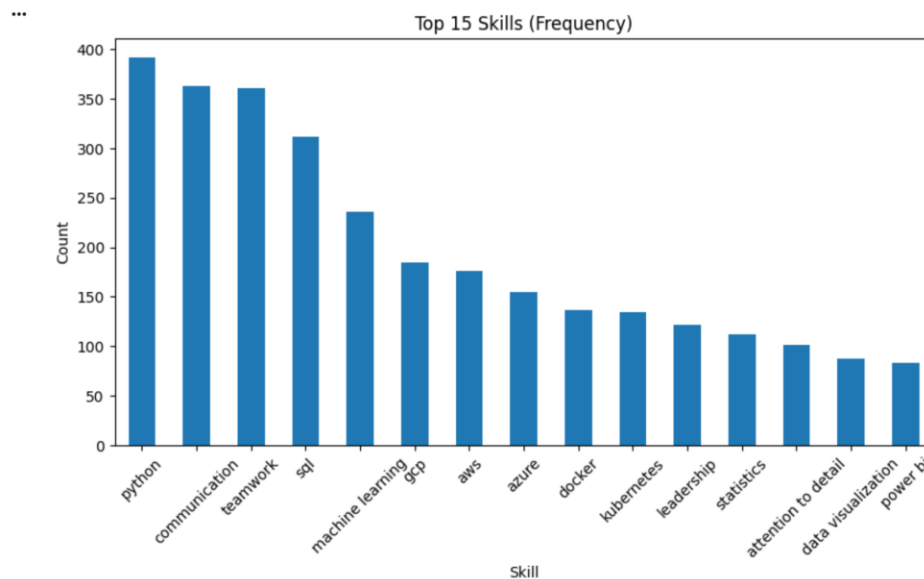
7.1.3 Skill Frequency Analysis

Which Skills are Most in Demand?

The results show a mix of technical tools and soft skills that employers in Saudi Arabia are looking for:

- **Top Technical Skills:** Python is the most demanded skill, appearing in nearly 400 postings. This is followed by SQL, which is essential for data management, and Machine Learning.
- **The Importance of Soft Skills:** Interestingly, Communication and Teamwork ranked very high, appearing almost as often as Python. This shows that Saudi employers don't just want someone who can code; they want people who can work well in a professional team and explain their findings.
- **Cloud and Infrastructure:** Skills in cloud platforms like GCP, AWS, and Azure, as well as tools like Docker and Kubernetes, are also very common.

- **Analysis and Visualization:** Tools for data storytelling, such as Power BI and general Data Visualization skills, are also frequently requested to help businesses understand their data.



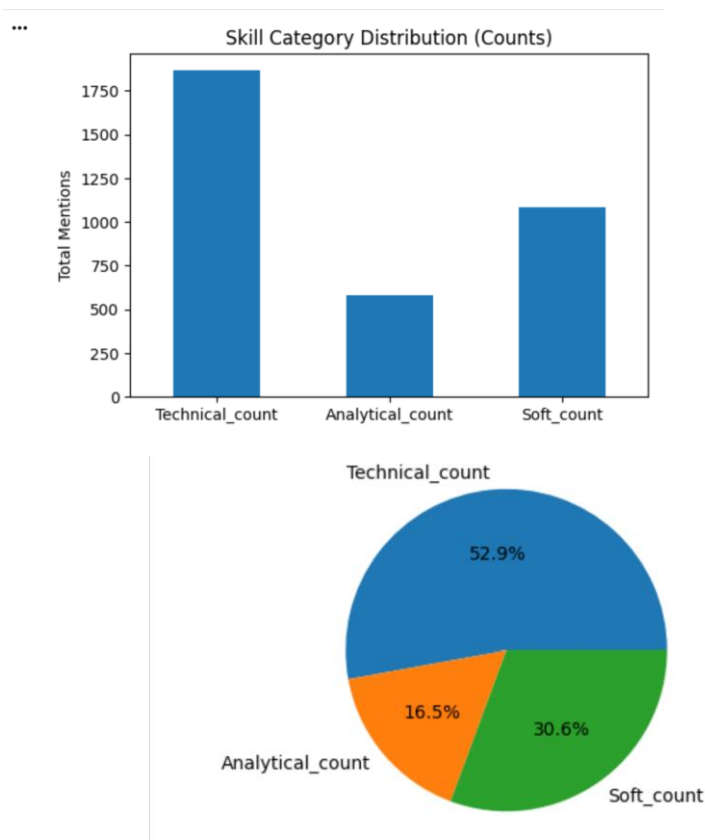
7.1.4 Skill Category Distribution

To get a better understanding of what employers prioritize, we grouped the identified skills into three main categories: Technical, Soft, and Analytical skills.

The results show a clear emphasis on practical ability:

- **Technical Skills (52.9%):** This is the largest category, accounting for more than half of all skill mentions. It includes the specific tools and programming languages like Python and SQL that are necessary to perform the job.
- **Soft Skills (30.6%):** soft skills make up nearly a third of the requirements. This confirms that being able to communicate and work in a team is just as important as technical knowledge.
- **Analytical Skills (16.5%):** While this is the smallest category, it still represents a significant portion of the demand, focusing on the ability to interpret data and use logic to solve problems.

Overall, the data suggests that while technical skills are essential for obtaining job opportunities, strong interpersonal skills play a significant role in helping candidates stand out in the hiring process



7.1.5 Role-Based Comparative Analysis

To see how skill requirements change between different jobs, we grouped the postings into categories like Data Scientist, Data Analyst and AI/ML Engineer. Even though many roles share common skills like Python or SQL, each has a very specific focus

1. Data Scientist vs Data Analyst

While both roles work with data, their tools are quite different:

- Data Analysts focus more on reporting and business intelligence. Their top skills include SQL , Power BI , and Excel
- Data Scientists are more focused on building models. Their most demanded skills are Machine Learning and Python , along with Statistics and Deep Learning

2. Data Engineer vs AI/ML Engineer

- Data Engineers are responsible for the infrastructure. They need SQL and Cloud platforms like AWS, GCP, and Azure to manage large data pipelines
- AI/ML Engineers focus on the actual AI implementation. Their top requirements are Machine Learning and specific frameworks like PyTorch and TensorFlow

3. Data Analyst vs. Data Engineer

Even though SQL is a top requirement for both roles, how they use it is very different :

- Data Analysts: They use SQL mainly for business intelligence. Their focus is on pulling data to create visualizations and reports that help managers make decisions.
- Data Engineers: For them, SQL is about building the actual system. They focus on data architecture, cloud performance, and making sure the data pipelines are fast and reliable.

This finding highlights that possessing SQL skills alone may not be sufficient. Instead, candidates are expected to understand how to apply SQL in ways that align with the specific requirements of different roles.

An important observation is that communication and teamwork skills appear among the top ten skills for nearly all roles. This suggests that, regardless of how technical the position may be, the ability to collaborate effectively with others remains a key requirement in the Saudi job market

ROLE: Data Scientist | postings: 66

count	
skills_extracted	
machine learning	49
python	43
statistics	40
sql	25
communication	23
teamwork	18
deep learning	18
nlp	17
pytorch	16
data visualization	14

ROLE: Data Analyst | postings: 84

count	
skills_extracted	
sql	45
power bi	38
communication	36
excel	33
statistics	29
python	27
attention to detail	26
tableau	24
data visualization	23
teamwork	22

ROLE: Data Engineer | postings: 96

count	
skills_extracted	
sql	63
python	47
teamwork	43
aws	26
machine learning	26
gcp	25
communication	23
azure	22
spark	18
power bi	18

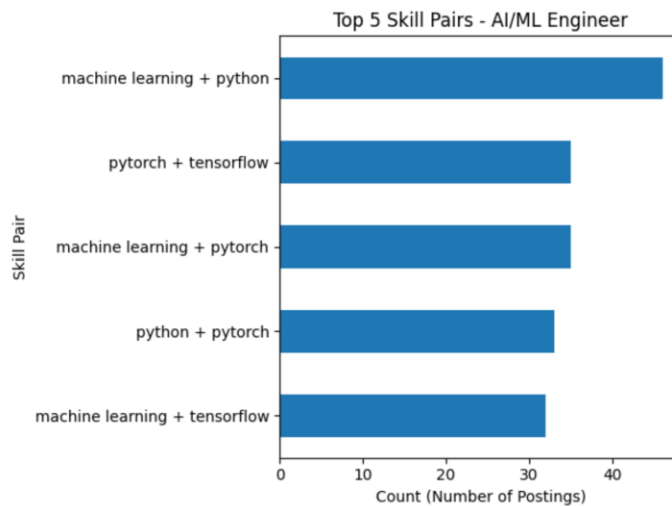
ROLE: AI/ML Engineer | postings: 113

count	
skills_extracted	
machine learning	70
python	69
pytorch	43
tensorflow	39
teamwork	38
communication	35
nlp	30
gcp	28
deep learning	24
aws	22

7.1.6 Skill Co-Occurrence Analysis

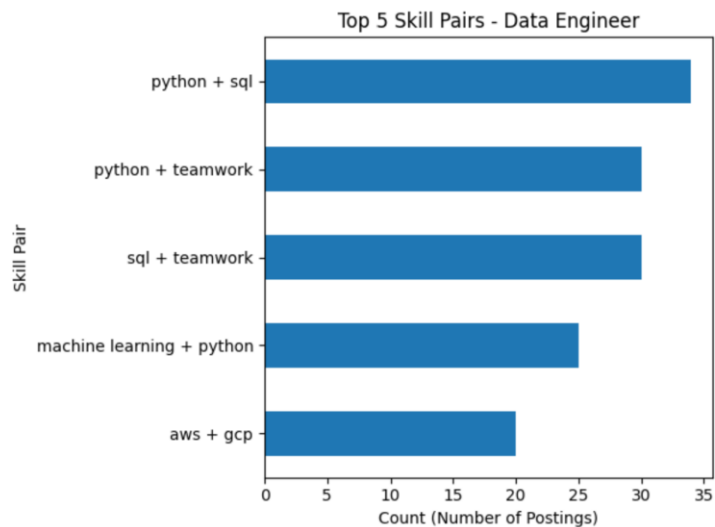
In addition to looking at individual skills, we analyzed which skills are most frequently requested together

For AI roles, the most frequent pairing is Machine Learning + Python, appearing in nearly 50 postings. This is closely followed by combinations of deep learning frameworks like PyTorch + TensorFlow. This indicates that employers expect AI candidates to be proficient in multiple frameworks rather than just one

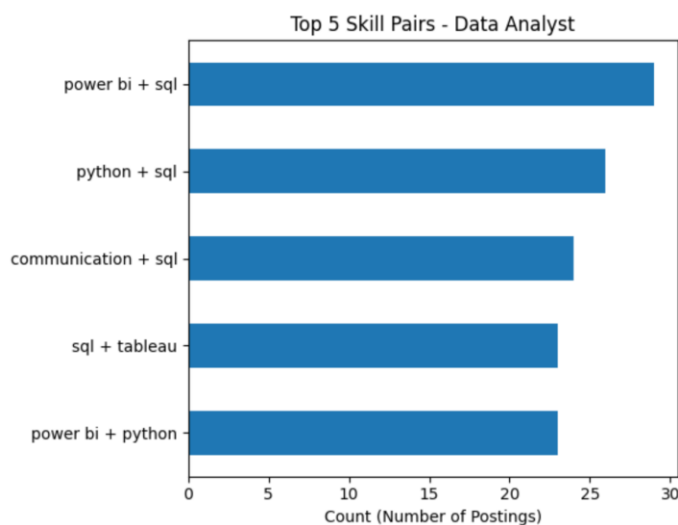


the strongest pair is **Python + SQL**, which makes sense as these are the core tools for data pipeline construction. We also see a high frequency of **Python + Teamwork** and **SQL + Teamwork**, which highlights that engineering is a

collaborative process. Another important technical bundle is **AWS + GCP**, suggesting that many companies are using multi-cloud environments.

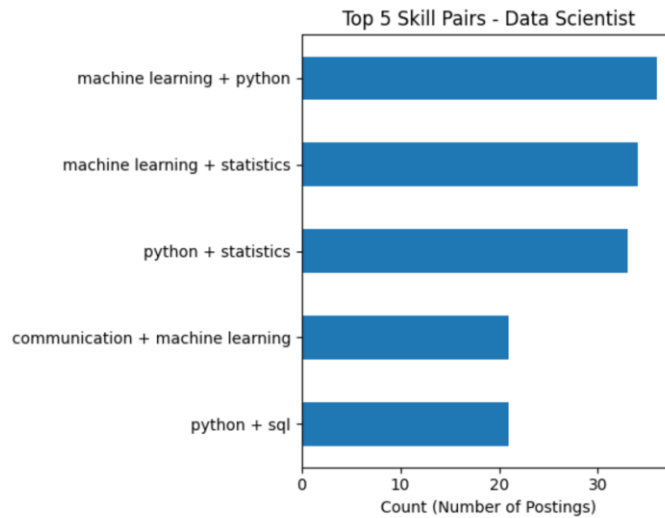


For Data Analysts, the most common combination is **Power BI + SQL**, appearing about 30 times. Other frequent pairs include **Python + SQL** and **SQL + Tableau**. This confirms that the ability to extract data (SQL) and visualize it (Power BI, Tableau) is the primary bundle for this role.



Data Science postings strongly link **Machine Learning + Python** and **Machine Learning + Statistics**, we also see a high co-occurrence of **Communication +**

Machine Learning. This suggests that for Data Scientists, the ability to explain complex models to others is just as important as the ability to build them



8. Secondary Data Analysis

To complement the analysis of the primary dataset, exploratory data analysis (EDA) was conducted on a secondary dataset obtained from Kaggle. The purpose of this analysis is to identify patterns in job roles, seniority levels, required skills, and job locations within the broader data science job market. By examining these distributions, the study aims to provide contextual insights that can be compared with the findings from the primary dataset.

8.1 Dataset Description

The secondary dataset used in this study was obtained from Kaggle [5]. The dataset contains 944 job postings related to data science roles, including positions such as Data Scientist, Machine Learning Engineer, and Data Engineer. It includes 13 variables, such

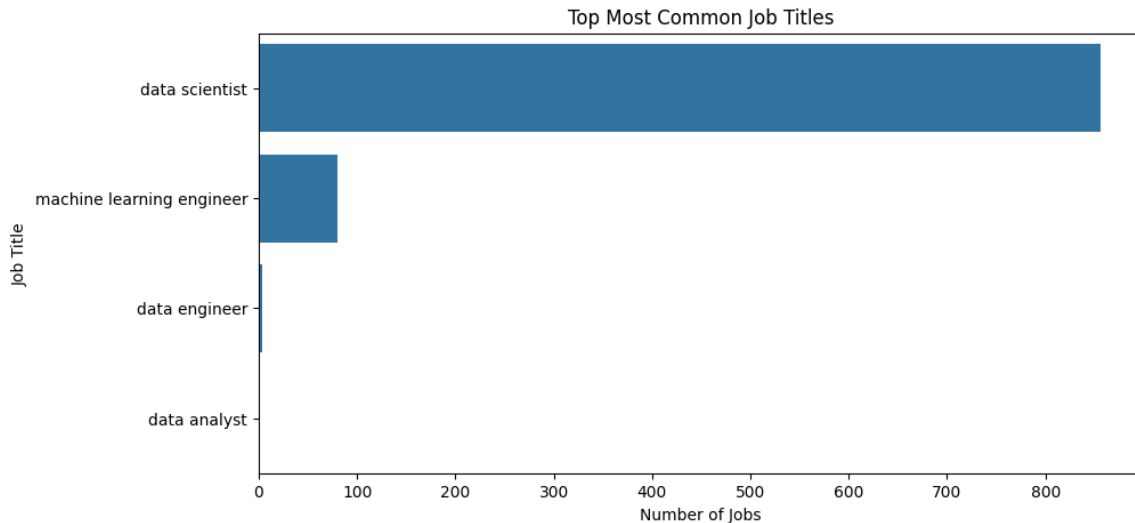
as job title, seniority level, company information, location, industry, salary, and required technical skills.

This dataset provides insight into the global demand for data science skills and roles, allowing us to compare patterns observed in the primary dataset with broader trends in the international job market.

8.2 Secondary Data Findings

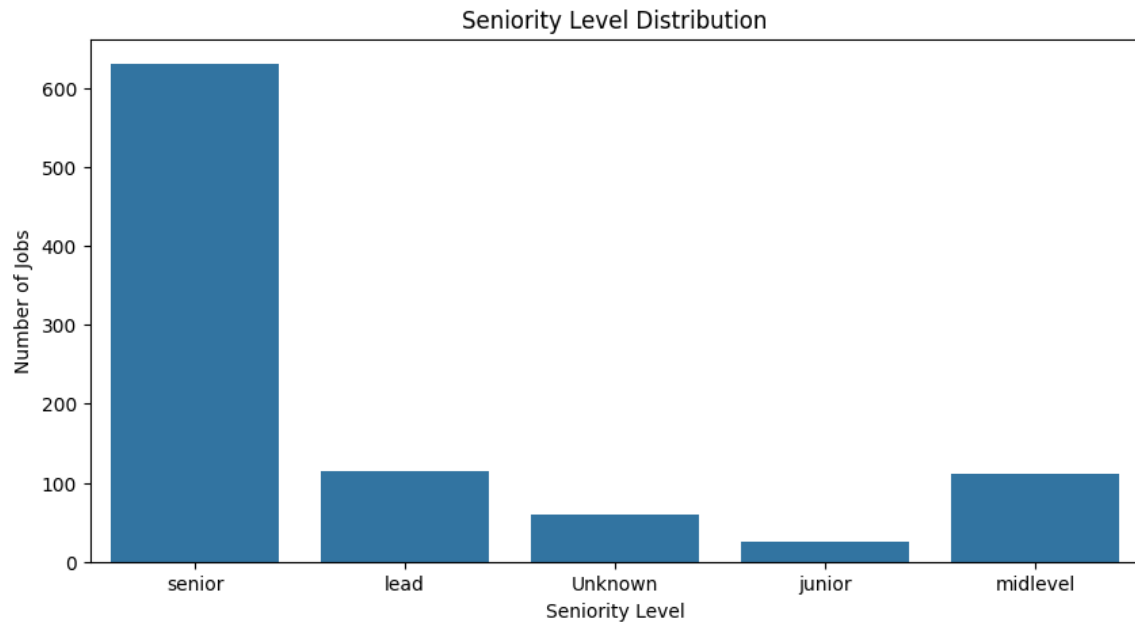
To better understand the characteristics of the secondary dataset, exploratory data analysis (EDA) was conducted to examine the distribution of job titles, seniority levels, required skills, and job locations.

8.2.1 Job Title Distribution



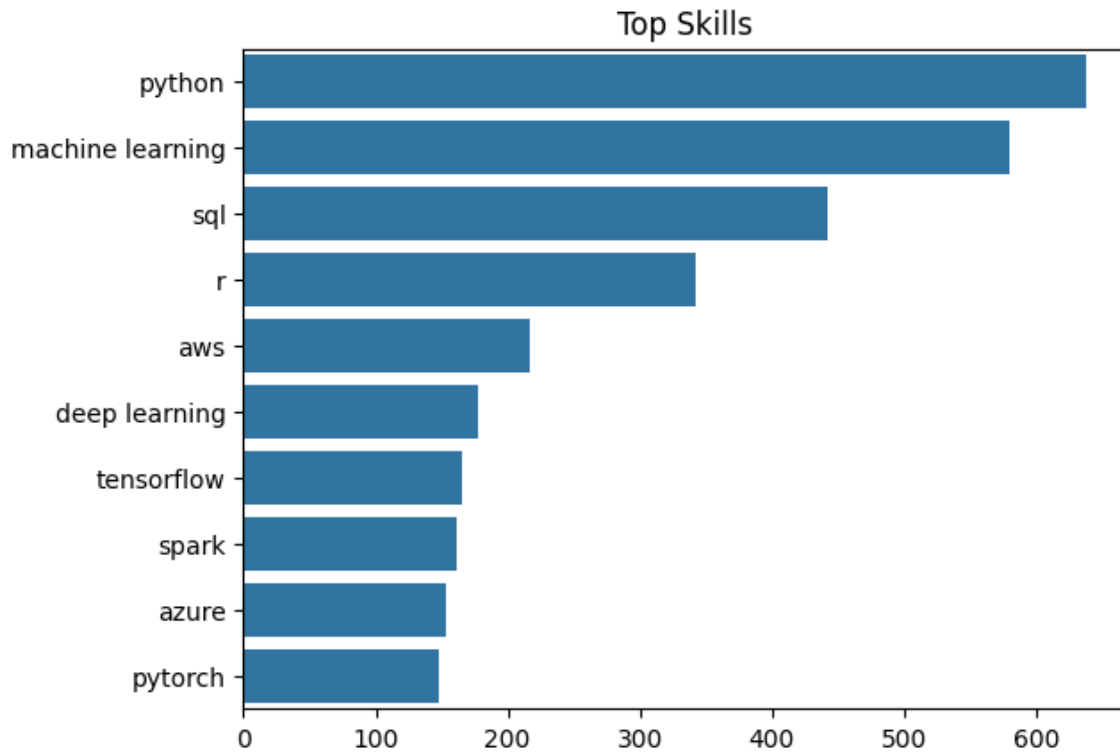
The distribution of job titles in the secondary dataset shows that Data Scientist roles dominate the dataset, representing the vast majority of job postings. Machine Learning Engineer positions appear as the second most common role, while Data Engineer and Data Analyst roles occur far less frequently. This suggests that the dataset focuses heavily on advanced data science roles rather than broader data-related positions. The dominance of Data Scientist roles indicates that organizations place strong emphasis on professionals capable of building models, analyzing large datasets, and supporting data-driven decision making.

8.2.2 Seniority Level Distribution



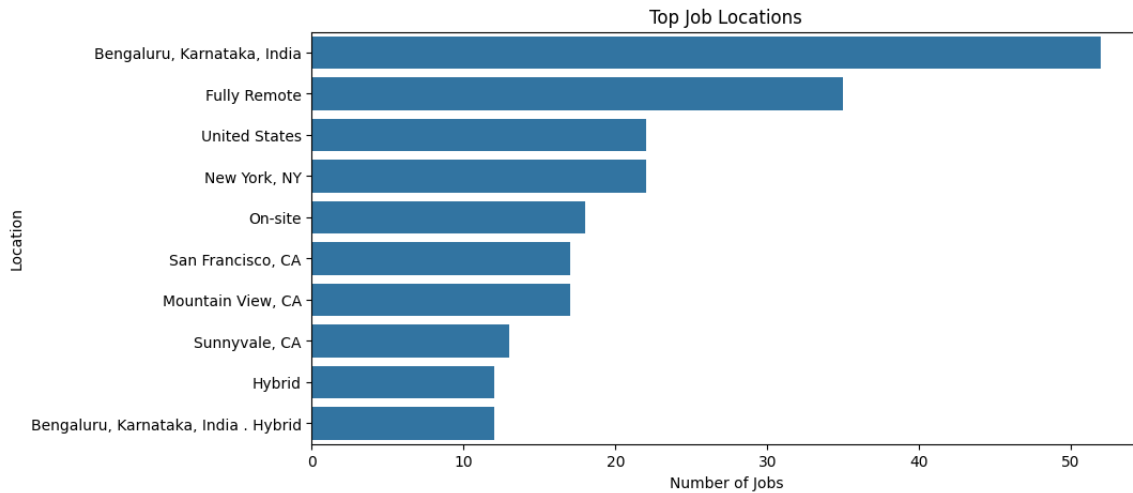
Most job postings target senior-level professionals rather than mid-level and lead roles. This pattern suggests that employers often seek experienced professionals with strong technical backgrounds and practical experience in data science and machine learning. The limited number of entry-level roles may indicate that organizations prioritize hiring candidates who can immediately contribute to advanced analytical tasks and model development.

8.2.3 Skills Analysis



The analysis of required skills shows that machine learning, Python, SQL, and R are among the most frequently mentioned technical skills in the dataset. Python appears particularly dominant, reflecting its widespread use in data analysis, machine learning, and artificial intelligence development. Cloud-related technologies such as AWS and machine learning frameworks like TensorFlow also appear among the common skills, highlighting the importance of scalable computing environments and advanced model development tools. These findings suggest that modern data science roles require a combination of programming, statistical analysis, and machine learning expertise.

8.2.4 Location Distribution



The geographic distribution of job postings shows that many positions are concentrated in major technology hubs such as Bengaluru, New York, and San Francisco. A significant number of postings also offer remote work, reflecting the growing adoption of distributed work environments in the technology sector. This pattern highlights the global demand for data science professionals, with opportunities concentrated in key tech centers while expanding through remote roles.

8.2.5 Limitations of Secondary Data

The secondary dataset has several limitations that should be considered when interpreting the results. The dataset represents global job postings instead of a country's job postings, which may limit direct comparisons with the primary dataset used in this study. Additionally, some variables contain missing values, such as seniority level, status, and revenue, which may affect the completeness of the analysis. Despite these limitations, the dataset still provides useful insights into broader trends in data science roles and skill demand.

9. Comparison Between Primary and Secondary Data

9.1 Compare Key Metrics

To compare the **primary dataset (jobs_sa_cleaned)** and the **secondary dataset (data_science_job_posts_2025)**, we focused on a common variable available in both sources: **skills**. Specifically, the comparison includes:

1. **Skills per job posting**
2. **Top skills frequency** (using count and share). The results are presented **side by side** to clearly identify significant differences.

1) Summary Statistics: Skills per Job

For each dataset, we computed the number of skills per job posting and summarized it using descriptive statistics (mean, median, quartiles, min/max).

Table 1 , Summary Statistics (Skills per Job)

Metric	Primary: jobs_sa_cleaned	Secondary: data_science_job_posts_2025
Number of job postings (N)	1020	944
Mean skills per job	3.46	4.43
Median skills per job	3.00	4.00
Std (skills count)	3.12	3.64
Q1 (25%)	1.00	1.00
Q3 (75%)	5.00	7.00
Min	0	0
Max	15	17
% postings with 0 skills	16.7%	21.3%

The secondary dataset has a higher mean and median, indicating that job postings in the secondary source tend to list more skills. Both datasets include postings with zero extracted skills, which suggests some missing or unreported skill information.

2) Top Skills: Frequency and Share

For each skill, we calculated:

- **count:** number of postings mentioning the skill
- **share:** proportion of postings mentioning the skill (count / N)

Top Skills, Primary (Saudi) (Top 10):

1. python (38.3%)
2. communication (35.6%)
3. teamwork (35.4%)
4. sql (30.5%)
5. machine learning (23.1%)
6. gcp (18.1%)
7. aws (17.3%)
8. azure (15.2%)
9. docker (13.4%)
10. kubernetes (13.1%)

Top Skills, Secondary (2025) (Top 10):

1. python (67.8%)
2. machine learning (61.4%)
3. sql (46.8%)
4. r (36.3%)
5. aws (23.1%)
6. deep learning (18.9%)
7. tensorflow (17.5%)
8. spark (17.1%)
9. azure (16.4%)
10. pytorch (15.7%)

3) Side-by-Side Comparison of Common Skills

This table satisfies the requirement to create side-by-side tables for spotting differences.

Table 2, Common Skills (Primary vs Secondary)

Skill	Primary share	Secondary share	Difference (Secondary – Primary)
python	38.3%	67.8%	+29.5%
sql	30.5%	46.8%	+16.3%
machine learning	23.1%	61.4%	+38.3%
aws	17.3%	23.1%	+5.8%
azure	15.2%	16.4%	+1.2%
gcp	18.1%	11.2%	–6.9%

9.2 Contextualize Findings

1) Similarities (Validation)

Python and SQL appear among the most frequent skills in both datasets. This consistency suggests that these are foundational requirements for data-related roles across different markets. The overlap supports the validity of the patterns observed in the Saudi (primary) dataset.

2) Differences

The secondary dataset shows a substantially stronger emphasis on machine-learning-related skills (e.g., Machine Learning, R, TensorFlow, PyTorch, Spark). In contrast, the primary dataset shows relatively higher visibility of cloud/engineering tools (e.g., GCP) and soft skills (e.g., communication and teamwork). These differences may reflect variations in market needs and the types of roles represented in each dataset.

3) Potential Reasons for Contradictions / Differences

If some findings appear to contradict between the datasets, plausible explanations include:

1. **Geographical differences:** the primary dataset reflects Saudi job postings, while the secondary dataset reflects a broader/different market.
2. **Sampling/platform differences:** job sources and platforms may differ, which affects the mix of companies and roles captured.
3. **Time-frame differences:** the secondary dataset is labeled as 2025 and may reflect more recent trends.
4. **Role composition (job-title mix):** different proportions of Analyst/BI vs DS/ML/Engineering roles lead to different skill frequencies.
5. **Skill reporting/extraction:** differences in how job ads list skills, and how skills were extracted/cleaned, can influence measured counts and shares.

4) Conclusion

Overall, the comparison indicates that core skills (Python and SQL) are consistently demanded in both datasets, while advanced DS/ML skills appear more frequently in the secondary dataset. This comparison provides broader context for interpreting the Saudi job-market findings and explains why certain skills may be more prominent in one dataset than the other.

10. Summary of New Insights and Hypotheses

10.1 New Insight

- **Python and SQL are “core skills” in both datasets.**
In both the Saudi dataset (primary) and the 2025 dataset (secondary), Python and SQL appear among the most common skills. This suggests these skills are a basic requirement for many data-related jobs.
- **The secondary dataset focuses more on advanced Data Science / ML skills.**
Skills like Machine Learning and tools such as R, TensorFlow, PyTorch, and Spark are more visible in the secondary dataset. This may mean the secondary dataset includes more DS/ML-heavy roles.
- **The Saudi dataset shows more cloud/engineering tools and soft skills.**
In the primary dataset, we see more cloud and engineering tools (like AWS, Azure, GCP, Docker, Kubernetes) and also soft skills (like communication and

teamwork). This suggests Saudi job posts may emphasize practical deployment skills and workplace skills more often.

- **Not all job posts list skills clearly.**

Both datasets include job posts where zero skills were extracted. This shows that some postings do not mention skills directly, or that skill extraction can miss some terms.

10.2 Hypotheses

- **H1: Cloud and engineering skills are relatively more important in Saudi job posts.**

We expect skills related to cloud and deployment (AWS/Azure/GCP, Docker, Kubernetes) to appear more often in the Saudi dataset than in the secondary dataset.

- **H2: The secondary dataset contains more specialized DS/ML roles.**

Because ML and deep learning tools appear more frequently in the secondary dataset, we expect it includes more Data Scientist / ML Engineer job types compared to the Saudi dataset.

- **H3: Differences in skills are partly caused by different job-title mixes.**

If one dataset has more analyst/Bi roles, we will see more reporting tools and soft skills. If another has more DS/ML roles, we will see more ML/DL frameworks. So the skill differences may reflect role distribution, not only market demand.

- **H4: Improving skill extraction will change the results.**

If we expand the keyword list and improve normalization (handling synonyms and different spellings), the number of extracted skills per job may increase, and the top-skill ranking may shift, especially in the primary dataset.

11. Modelling and Results

The goal is to build and evaluate classification models that can predict the job role based on the skills listed in job postings. The problem is formulated as a classification task, where each input is assigned to one of four predefined roles: **Data Scientist, Data Analyst, Data Engineer, or AI/ML Engineer.**

11.1 Models Used

Three classification models were implemented and compared :

- Logistic Regression: used as a baseline model due to its simplicity and effectiveness with high-dimensional text data such as TF-IDF features.
- Decision Tree: used to explore non-linear relationships between skills and job roles by learning rule-based patterns.
- Random Forest: used as an ensemble method that combines multiple decision trees to improve performance and reduce overfitting.

These models were selected to provide a balanced comparison between simple, interpretable models and more complex models.

11.2 Results

The performance of the models was evaluated using Accuracy and F1-score. The results are summarized below:

- Logistic Regression:

Accuracy = 93.18%

F1 = 0.9310

- Decision Tree:

Accuracy = 90.91%

F1 = 0.9087

- Random Forest:

Accuracy = 94.32%

F1 = 0.9416

All models achieved strong performance, with accuracy above 90%, indicating that the extracted features are effective in distinguishing between job roles

11.3 Best Model Selection

Random Forest achieved the highest performance among the three models, with an accuracy of 94.32% and the highest F1-score

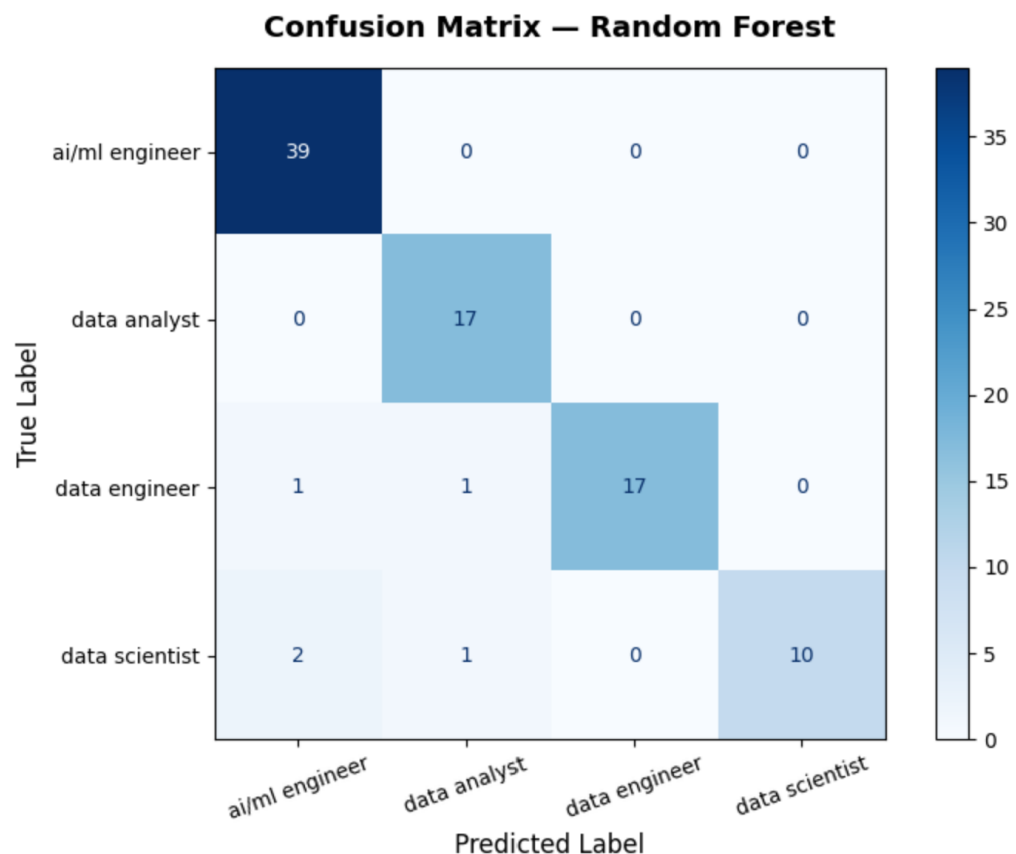
Therefore, Random Forest was selected as the final model for this task. The improvement over Logistic Regression is small but consistent, indicating that the ensemble approach provides a slight advantage in capturing more complex patterns in the data

11.4 Interpretation of Results

The results show that the models are able to distinguish between job roles effectively based on the skills listed in job postings. The strong performance across all models suggests that the TF-IDF representation successfully captures meaningful differences between roles.

However, some misclassifications still occur, particularly between roles with similar skill requirements, such as Data Scientist and AI/ML Engineer. This is expected because these roles often share similar skills in real job postings.

Overall, the results indicate that while the models perform well, some confusion comes from the data itself rather than the model.



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